



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

17-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE-II)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	A	C	B	A	B	B	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	D	C	B	A	D	C	D	B

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	D	C	D	B	B	B	D	D
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	B	C	D	B	B	D	C	D

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	A	D	A	B	B	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	D	C	D	D	C	B	C	A

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (C)**

$$n_{fo} = 221$$

$$m_{fo} = 238$$

$$\frac{n}{m} = \frac{13}{14} \Rightarrow n = 13$$

$$\therefore f_o = \frac{221}{13} = 17$$

2. **Ans (B)**

$$\Delta\lambda = \frac{V}{C}\lambda$$

$$\Delta\lambda = \frac{2 \times 10^6 \times (3000)}{3 \times 10^8} = 20 \text{ \AA}$$

3. **Ans (D)**

$$f = f_l \sqrt{\frac{3+1}{1}} = 2:1 = \frac{\lambda_2}{\lambda_1}$$

4. **Ans (A)**

$$A \propto e^{-bt/2m}$$

$$A_1 = \frac{A}{3} \propto (e^{-b/2m})^3$$

$$A_2 \propto (e^{-b/2m})^4$$

$$A_2 = \left(\frac{1}{3}\right)^4 A_0$$

5. **Ans (C)**

$$y = A_0 + \sqrt{3}A \sin(\omega t + \phi)$$

$$\text{Amplitude } \sqrt{3}A$$

6. **Ans (B)**

$$\frac{T}{3} = \frac{1.5}{3} = 0.5$$

7. **Ans (A)**

$$\frac{x^2}{A^2} + \frac{v^2}{(A\omega)^2} = 1$$

$$\frac{A}{A\omega} = \frac{8}{10}$$

$$\omega = \frac{5}{4}$$

$$\frac{2\pi}{T} = \frac{5}{4}$$

$$T = \frac{8\pi}{5}$$

8. **Ans (B)**

$$\Delta Q = \Delta U + \Delta W$$

$$nC\Delta T = nC_v\Delta T + \frac{nR\Delta T}{1-x}$$

$$C = \frac{3}{2}R + \frac{R}{1-x}$$

$$C = \frac{3}{2}R + 3R = \frac{9R}{2}$$

$$(PV = nRT$$

$$PT^2 = K$$

$$P^3V^2 = K$$

$$PV^{2/3} = K)$$

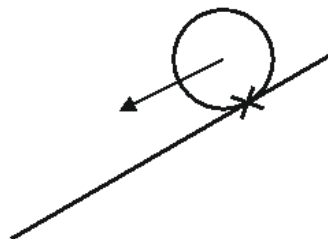
9. **Ans (B)**

$$K = \frac{Ay}{\ell}$$

$$\frac{U_1}{U_2} = \frac{K_2}{K_1} = \frac{1}{2}$$

$$U = \frac{1}{2}Kx^2 = \frac{F^2}{2K}$$

10. **Ans (B)**



$$(mg \sin \theta) R = \frac{3}{2}mR^2 \left(\frac{a}{R} \right)$$

$$\frac{2}{3}g \sin \theta = a$$

$$S = \frac{1}{2}at^2$$

$$t \propto \frac{1}{\sqrt{a}}$$

$$a_1 = g \sin \theta$$

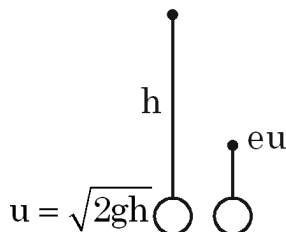
$$a_2 = \frac{2}{3}g \sin \theta$$

$$\frac{t_1}{t_2} = \sqrt{\frac{a_2}{a_1}} = \sqrt{\frac{2}{3}}$$

11. **Ans (A)**

$$U = mgh$$

12. **Ans (C)**



$$u = \sqrt{2gh}$$

$$h + \frac{(eu)^2}{2g} + \frac{e^4u^2}{2g} + \dots$$

$$h + \frac{2e^2u^2}{2g}(1 + e^2 + e^4 + \dots)$$

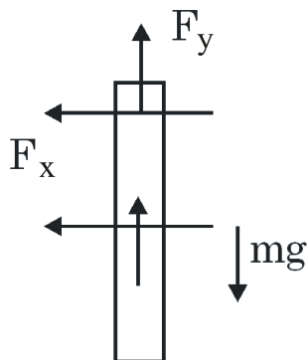
$$h + \frac{2e^2}{2g}(2gh) \left(\frac{1}{1-e^2} \right)$$

$$= \frac{h - e^2h + 2e^2h}{1-e^2}$$

$$\frac{h(1+e^2)}{(1-e^2)} = H_{\text{total}}$$

$$\frac{(1+\frac{1}{4})}{(1-\frac{1}{4})} = \frac{5}{3} \times 1 = 15m$$

13. Ans (D)



$$F_x = \frac{m\ell}{2}\alpha = 0 \quad \dots(1)$$

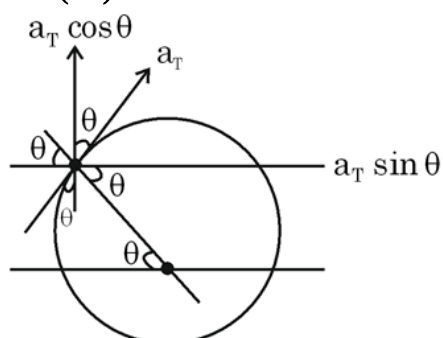
$$F_y - Mg = \left(M\frac{\ell}{2}\right)\omega^2 \quad \dots(2)$$

$$mg\frac{\ell}{2} = \frac{1}{2}\left(\frac{m\ell^2}{3}\right)\omega^2$$

$$\frac{3Mg}{2} = \frac{m\ell\omega^2}{2}$$

$$\therefore F_y = \frac{5}{2}mg = 75\text{N}$$

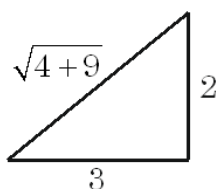
14. Ans (C)



$$a_T \sin \theta = 2$$

$$a_T \cos \theta = 3$$

$$\tan \theta = \frac{2}{3}$$



$$\cos \theta = \frac{3}{\sqrt{13}}$$

15. Ans (B)

$$\frac{1}{5}N_0 = N_2 \left(\frac{4}{5} \text{ decayed}\right)$$

$$\frac{4}{5}N_0 = N_1$$

$$N_2 = \frac{N_1}{2^n} \Rightarrow n = 2$$

16. Ans (A)

$$L = f_0 + f_e$$

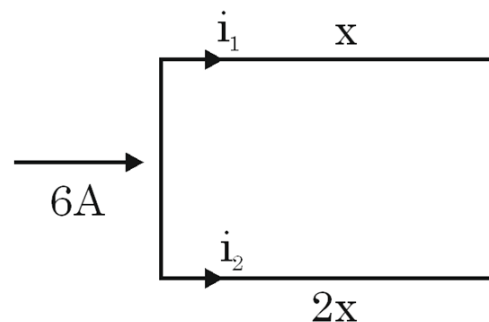
$$M = \frac{f_0}{f_e} = 15$$

$$f_0 = 15f_e$$

$$\therefore 80 = 16f_e$$

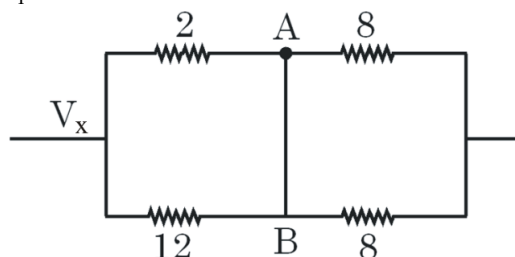
$$5 = f_e$$

17. Ans (D)



$$i_1 = \frac{2}{3} \times 6 = 4\text{A}$$

$$i_1 = 2\text{A}$$



$$V_x - V_A = 4 \times 2$$

$$V_x - V_B = 2 \times 12$$

$$V_A - V_B = 24 - 8 = 16$$

18. Ans (C)

$$1 = x \frac{\frac{1}{2}(1) \left(V^2 - \frac{1}{2}(6) \left(\frac{V^2}{36}\right)\right)}{\frac{1}{2}(1)(V^2)}$$

$$= \left(1 - \frac{1}{6}\right) \times 100$$

$$= \frac{5}{6} \times 100\% = 83.33\%$$

19. Ans (D)

$$L = \mu_0(n)^2 A \ell$$

$$n^1 = 2n$$

A same

$$\ell = 2\ell,$$

$$L^1 = 8L = 24\text{ H}$$

20. **Ans (B)**

$$R_1 + 20 = R_2 \quad \dots(1)$$

$$\frac{45}{R_1} = \frac{55}{R_2} \quad \dots(2)$$

$$R_2 = \frac{11}{9} R_1$$

Put in (1)

$$20 = \frac{2}{9} R_1$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (26)**

$$\frac{x}{100} = \frac{67 - 15}{200} = \frac{52}{200}$$

$$x = 26$$

2. **Ans (504)**

$$s = \frac{t^4}{2}$$

$$v = \frac{ds}{dt} = 2t^3$$

$$w = k_f - k_i$$

$$= \frac{1}{2} m (v_f^2 - v_i^2)$$

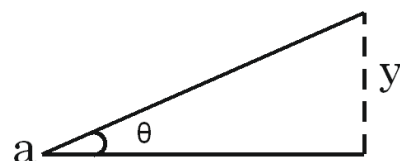
$$= 2 (4) (8^2 - 1) = 504$$

3. **Ans (1248)**

MSR + CSR (L.C)

$$= 12 + \frac{24 \times 1}{50} = 12.48$$

4. **Ans (92)**



$$a \sin \theta = n \lambda$$

$$\frac{ay}{D} = n \lambda$$

$$y = \frac{n \lambda D}{a}$$

$$y_2 - y_1 = \frac{\lambda D}{a} (3 - 1) = 0.0092$$

5. **Ans (13)**

$$i = \sqrt{i_R^2 + (i_L - i_C)^2} = 13$$

6. **Ans (1999)**

$$60\pi \times 10^{-6} = B(30 \times 10^{-6})$$

$$2\pi = B = \mu_0 \mu_r H$$

$$\frac{2\pi}{4\pi \times 10^{-7} \times 2500} = \mu_r = 1 + x$$

$$x = 1999$$

7. **Ans (8)**

$$V_T = \varepsilon - ir$$

$$\varepsilon = 48, r = \frac{\varepsilon}{i} = 8$$

8. **Ans (12)**

$$\varepsilon_{\max} = N B A \omega$$

$$i_{\max} = \frac{N B A \omega}{R}$$

$$i_{\text{rms}} = \frac{i_{\max}}{\sqrt{2}}$$

$$= \frac{100 \times 5 \times 10^{-4} \times \pi (4) 2\pi \times 30\sqrt{2}}{\sqrt{2} (\pi^2)} = 12$$

9. **Ans (32)**

$$\frac{1}{\lambda_2} = \left(1 - \frac{1}{9}\right) = \frac{8}{9}$$

$$\frac{1}{\lambda_1} = \left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5}{4 \times 9}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{8 \times 4 \times 9}{9 \times 5} = \frac{32}{5}$$

10. **Ans (4)**

$$\frac{1}{2} K \left(\frac{A^2}{4} \right) = U$$

$$K \varepsilon = \frac{1}{2} K \left(A^2 - \frac{A^2}{4} \right) = 3 (U)$$

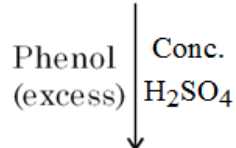
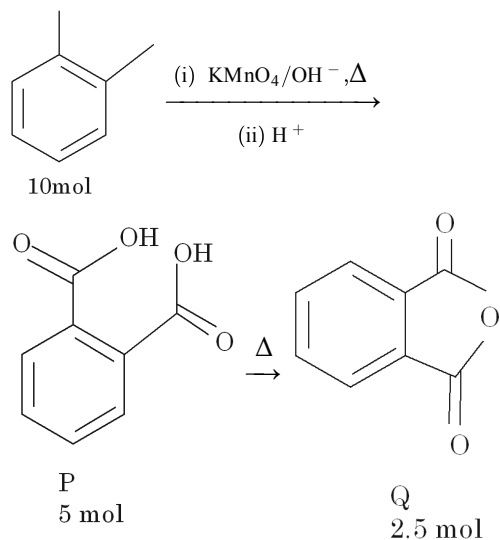
$$T = 2\pi \sqrt{LC}$$

$$\frac{T}{6} = 4 \times 10^{-3} \text{ s}$$

PART-2 : CHEMISTRY

SECTION-I

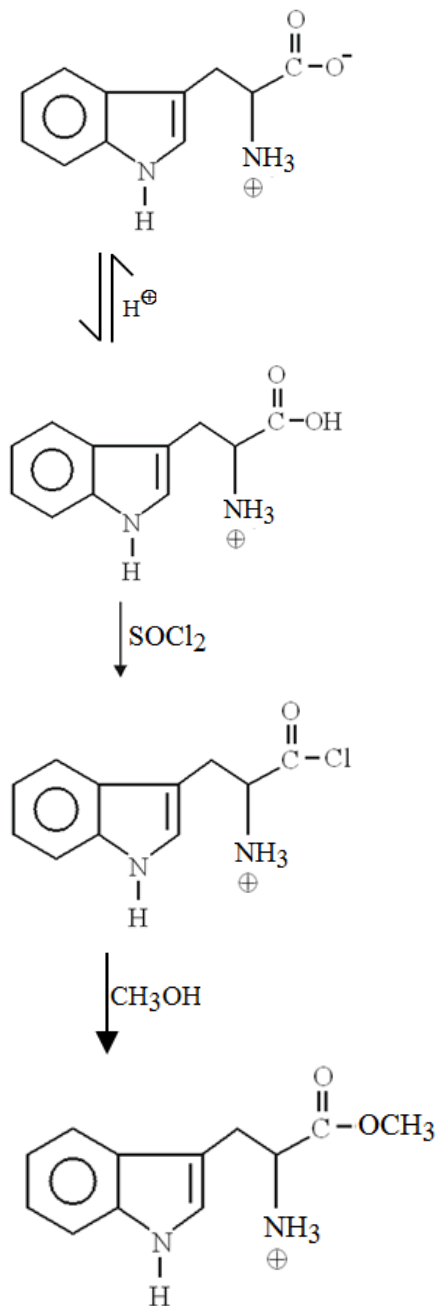
1. Ans (D)



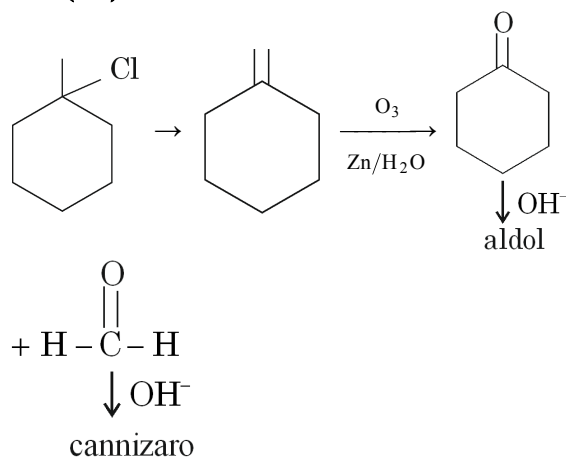
Phenolphthalein
2.5 mol

$$W = 2.5 \times 318 = 795.00 \text{ g}$$

2. Ans (B)



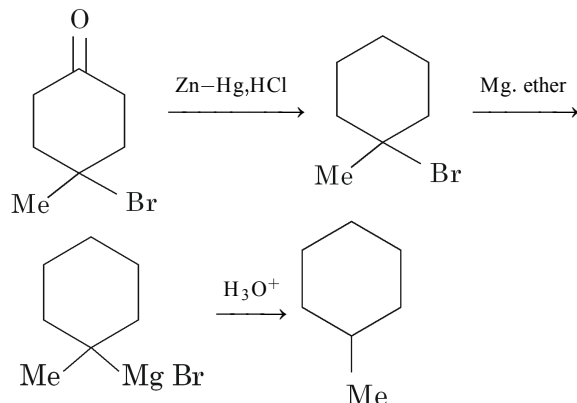
3. Ans (D)



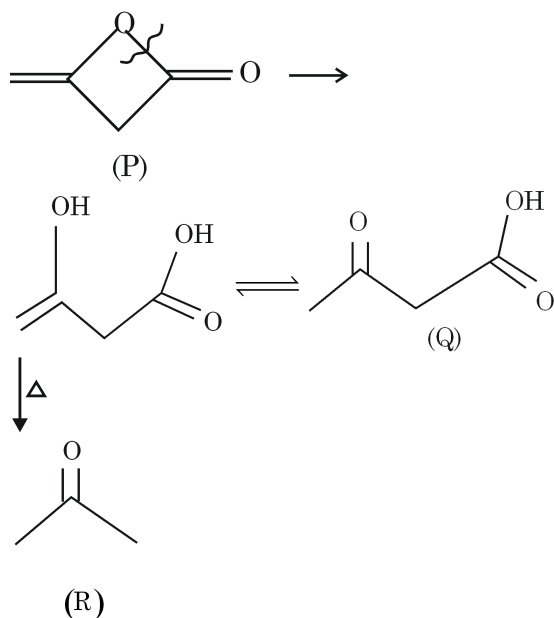
4. **Ans (C)**

B_2H_6 effects only acid.

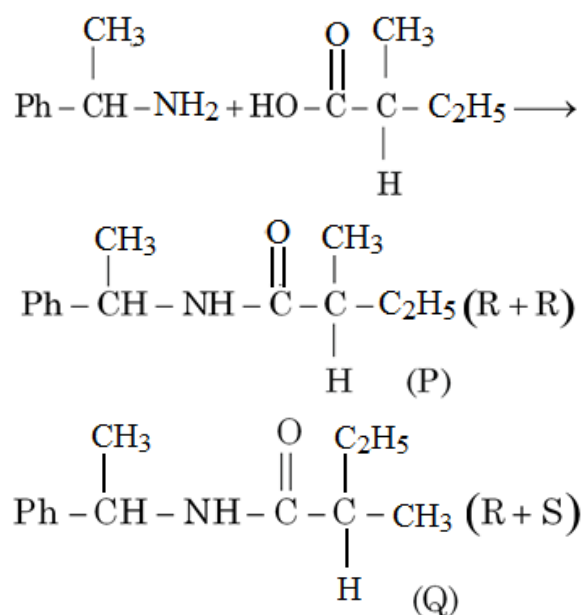
5. **Ans (D)**



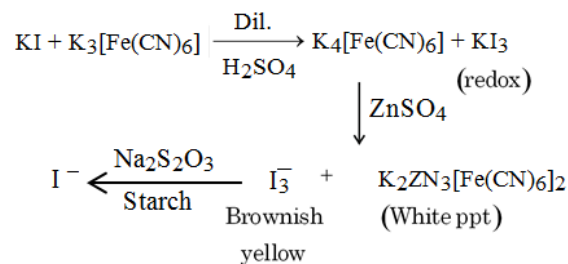
6. **Ans (B)**



7. **Ans (B)**

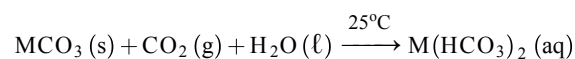
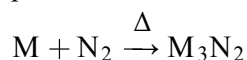


8. **Ans (B)**



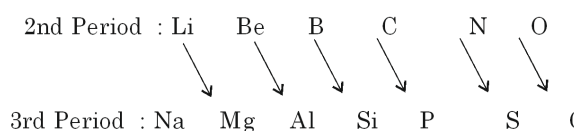
9. **Ans (D)**

Group-2 metals (Be, Mg, Ca, Sr, Ba) nitrides (M_3N_2) and their hydrogen bicarbonates ($M(HCO_3)_2$) are known only in solution phase

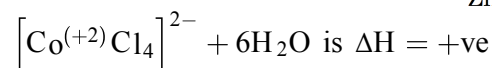
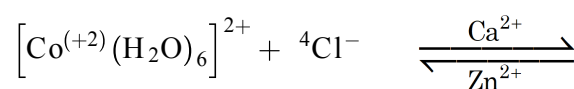


The above two properties are not given by group - I metals (Li, Na, K, Rb, Cs) except Li. Li resembles with group-II metals in comparison of their chemical properties.

Specially, Li(A), and Mg(B) have similar chemical properties as they show "disagonal relationship" in the periodic table.



10. **Ans (D)**



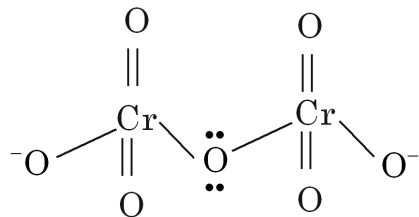
Ca^{2+} is class - a metal, Zn^{2+} is class - b metal.

Class - a metals form stable complexes with ligands N, O, F as donor atom, class - b metals form stable complexes with the ligands P, S, Cl as donor atoms.

12. **Ans (B)**

Major component of portland cement is "Tricalcium silicate (51%, $3CaO \cdot SiO_2$)

13. Ans (B)



dichromate ion contain non-linear symmetrical
Cr – O – Cr Bond

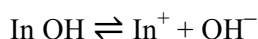
14. Ans (C)

$$\text{Efficiency} = \frac{\Delta G}{\Delta H} = \frac{-237.39}{-241.00} = 98.5\%$$

15. Ans (D)

$$\text{pH of NaH}_2\text{PO}_4 = \frac{pK_{a1} + pK_{a2}}{2} = \frac{7.2 + 10.6}{2} = 8.9$$

16. Ans (B)



$$K_b = \frac{[\text{In}^+][\text{OH}^-]}{[\text{In OH}]}$$

$$\text{Also } \text{pOH} = \text{p}K_b + \log \frac{[\text{In}^+]}{[\text{In OH}]}$$

At pH = 3.3, pOH = 14 - 3.3 = 10.7, the
indicator will be sufficient ionized to give the
colour solely due to In⁺ ion.

Hence,

$$\begin{aligned} 10.7 &= \text{p}K_b + \log \frac{[\text{In}^+]}{[\text{In OH}]} \\ &= (14 - \text{p}K_a) + \log \frac{[\text{In}^+]}{[\text{In OH}]} \\ &= \left[14 - \frac{(3.3 + 4.7)}{2} \right] + \log \frac{[\text{In}^+]}{[\text{In OH}]} \\ \Rightarrow \log \frac{[\text{In}^+]}{[\text{In OH}]} &= 0.7 = \log 5 \\ \Rightarrow \frac{[\text{In}^+]}{[\text{In OH}]} &= 5 \end{aligned}$$

17. Ans (B)



At t = 0 400 mm

After 2 hrs 400 - x x x x

As per question 400 - x + x + x + x = 800

$$\Rightarrow x = 200$$

$$\begin{aligned} k &= \frac{2.303}{2} \log \frac{P_A^0}{P_A} \\ &= \frac{2.303}{2} \log \frac{400}{200} \\ &= 0.346 \text{ hr}^{-1} \end{aligned}$$

18. Ans (D)

For a BCC lattice,

$$\frac{X^+}{Y^-} = 0.732 \Rightarrow Y^- = \frac{0.4}{0.732} = 0.546^\circ \text{A}$$

Now for fcc lattice,

$$\frac{Z^+}{Y^-} = 0.414 \Rightarrow Z^+ = 0.546 \times 0.414 = 0.226^\circ \text{A}$$

19. Ans (C)

$$\begin{aligned} Z &= \frac{P V_m}{RT} = \frac{V_m}{V_m - b} = \frac{11b}{11b - b} = \frac{11}{10} \\ \therefore \frac{P(V - nb)}{n} &= RT \Rightarrow \frac{RT}{P} = \frac{V - nb}{n} = V_m - b \end{aligned}$$

20. Ans (D)

$$\text{Moles of Sn}^{2+} = 523.15 \times 10^{-3}$$

$$\text{Moles of MnO}_4^- \text{ required} = \frac{2}{5} \times 523.15 \times 10^{-3}$$

Volume

$$= \frac{\frac{2}{5} \times 523.15 \times 10^{-3} \times 1000}{0.5} = 418.52 \text{ mL}$$

PART-2 : CHEMISTRY

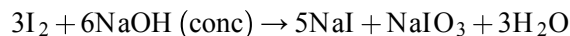
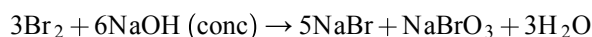
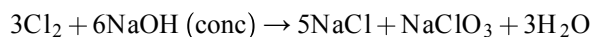
SECTION-II

1. Ans (2)

a, b, d, e are compounds which give Reimer
Tiemann and other do not give.

$$\text{So, } 4/2 = 2$$

2. **Ans (1)**

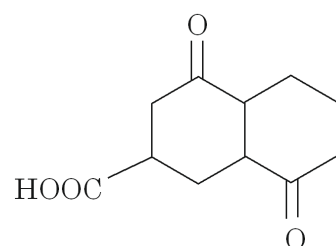
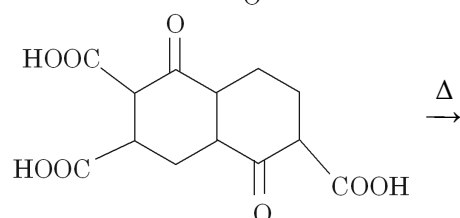
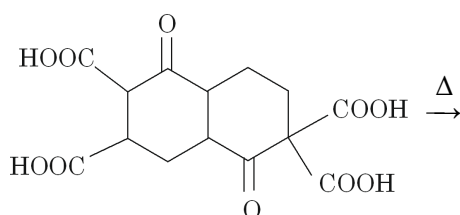


Total 6 compounds undergo disproportionation

So $x = 4$ & $y = 4$

$x / y = 1$

3. **Ans (3)**



4. **Ans (62)**

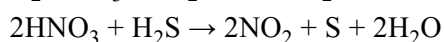
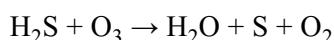
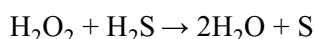
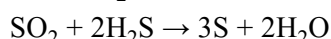
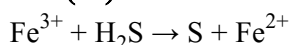
$$\Delta T = \frac{1000 \times K_b' \times w}{m \times W}$$

$$m_{\text{exp}} = 123.8$$

$$\frac{m_N}{m_{\text{exp}}} = 1 - \alpha + \frac{\alpha}{n}$$

$$\therefore \alpha = 1 \Rightarrow \frac{m_N}{m_{\text{exp}}} = \frac{\alpha}{n}$$

5. **Ans (5)**



6. **Ans (0)**

Acetylene gives Tollen's test but not form silver mirror (Ag). (Ag_2C_2 - white ppt)

7. **Ans (10)**

$$\frac{49 \times \frac{1}{5} \times 1000}{40} + \frac{49 \times \frac{4}{5} \times 1000}{49} = V \times \frac{1}{10}$$

$$V = 10450 \text{ ml} = 10.45 \text{ litre}$$

8. **Ans (10)**

$$\text{P.V.} = +4$$

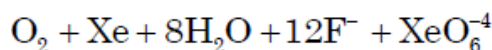
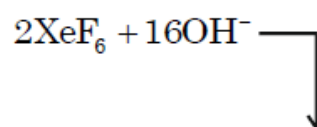
$$\text{S.V.} = 6$$

9. **Ans (5)**

All monosaccharides and Disaccharides (except sucrose) are Reducing sugars.

10. **Ans (8)**

$$(8 + 0)$$



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (D)**

$$n(S) = \frac{9!}{3! 2!} = 9.8.7.6.10$$

Now, for $n(E)$

Case 1 : No two T's together using GAP

$$\text{method for T's} \Rightarrow {}^7\text{C}_3 \cdot \frac{6!}{2!} = 12600$$

Case 2 : No two T's together and N's together

using GAP method for T's, keeping N's

together as a single object $\Rightarrow 5! \cdot {}^6\text{C}_3 = 2400$

$$\therefore n(E) = 12600 - 2400 = 10200$$

$$\therefore \text{Required pr} = \frac{n(E)}{n(S)} = \frac{10200}{9.8.7.6.10} = \frac{85}{252}$$

2. **Ans (A)**

$$\text{Since } 61\vec{p} + 40\vec{q} + 21\vec{r} = 0$$

$$\Rightarrow \vec{p}, \vec{q}, \vec{r} \text{ are coplanar}$$

$$\Rightarrow \vec{q} \times \vec{r} \text{ and } \vec{r} \times \vec{p} \text{ are collinear}$$

$$\Rightarrow (\vec{q} \times \vec{r}) \times (\vec{r} \times \vec{p}) = \vec{0}$$

3. **Ans (B)**

$$I_n = \int_0^1 (1-x^3)^n dx$$

$$= \left[(1-x^3)^3 \cdot x \right]_0^1 - \int_0^1 n(1-x^3)^{n-1} \cdot (-3x^2) x dx$$

$$= 0 - 3n \int_0^1 (1-x^3)^n dx + 3n \int_0^1 (1-x^3)^{n-1} dx$$

$$= -3nI_n + 3nI_{n-1}$$

$$\therefore I_n (1+3n) = 3nI_{n-1} \Rightarrow \frac{I_n}{I_{n-1}} = \frac{3n}{3n+1}$$

4. **Ans (A)**

$$I = \int_{\frac{1}{2018}}^{2018} \frac{\cos x}{x \left\{ \cos x + \cos \frac{1}{x} \right\}} dx ; \quad \text{put } x = \frac{1}{t}$$

$$dx = \frac{-1}{t^2} dt$$

$$\dots\dots\dots(1)$$

$$\therefore I = \int_{2018}^{\frac{1}{2018}} \frac{\frac{-1}{t^2} \cos \frac{1}{t}}{\frac{1}{t} \left\{ \cos \frac{1}{t} + \cos t \right\}} dt$$

$$= \int_{\frac{1}{2018}}^{2018} \frac{\cos \frac{1}{t}}{t \left\{ \cos t + \cos \frac{1}{t} \right\}} dt$$

$$= \int_{\frac{1}{2018}}^{2018} \frac{\cos \frac{1}{x}}{x \left\{ \cos x + \cos \frac{1}{x} \right\}} dx \dots\dots\dots(2)$$

$$(1) + (2) \Rightarrow 2I = \int_{\frac{1}{2018}}^{2018} \frac{1}{x} dx = [\ln x]_{\frac{1}{2018}}^{2018}$$

$$\therefore I = \ln 2018$$

5. **Ans (D)**

$$\therefore \frac{a}{b} R \frac{a}{b} \Rightarrow ab = ab \quad \therefore \text{Re flexive}$$

$$\therefore \frac{a}{b} R \frac{c}{d} \Rightarrow ad = bc \Rightarrow bc = ad \Rightarrow \frac{c}{d} R \frac{a}{b}$$

$$\therefore \text{symmetric}$$

$$\therefore \frac{a}{b} R \frac{c}{d} \Rightarrow ad = bc \text{ \& } \frac{c}{d} R \frac{e}{f} \Rightarrow cf = de$$

$$\Rightarrow adcf = bcde \Rightarrow af = be \Rightarrow \frac{a}{b} R \frac{e}{f}$$

$$\therefore \text{transitive}$$

6. **Ans (A)**

$$\bar{w} = a\bar{x} + b \Rightarrow 55 = 48a + b$$

$$\sigma_w^2 = a^2 \sigma_x^2$$

$$15^2 = a^2 \cdot 12^2 \Rightarrow a = \frac{5}{4}, \quad b = -5$$

7. **Ans (B)**

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{3, 5, 7\}; \quad U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$C = \{5, 10\}$$

$$n(A^C \cap B^C \cap C^C) = n(U) - n(A \cup B \cup C)$$

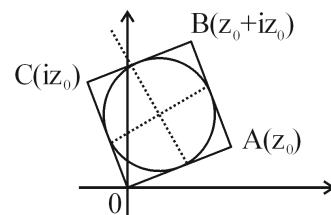
$$= 10 - 7 = 3$$

8. **Ans (B)**

$$\sum_{r=0}^{2020} {}^{2020}C_r (x-2018)^{2020-r} (2017)^r = (x-1)^{2020}$$

Hence coefficient of $x^{2017} = -({}^{2020}C_{2017})$

9. **Ans (B)**



Clearly mid-point of OB is centre of the circle and radius is equal $\frac{|z_0|}{2}$

$$\text{Required equation is } \left| z - \frac{z_0}{2}(1+i) \right| = \frac{|z_0|}{2}$$

10. **Ans (A)**

Negation of $\sim (p \vee q) = (p \vee q)$
 $\Rightarrow$ you want to succeed or you will find a way

11. **Ans (D)**

$$x^2 = e^{\frac{y}{x} \frac{dy}{dx}}$$

$$\int y dy = \int 2x \ln x dx$$

$$\frac{y^2}{2} = 2 \left(\frac{x^2 \ln x}{2} - \frac{x^2}{4} \right) + c$$

$$y^2 = x^2 (\ln x^2 - 1) + c$$

12. **Ans (B)**

$$\theta = 2\sec^{-1}(e)$$

$$\therefore 2\sec^{-1}(e) = \frac{\pi}{3}$$

$$\Rightarrow e = \frac{2}{\sqrt{3}}$$

13. **Ans (D)**

$$\therefore f(x-2) + f(x) + f(x+2) = 0 \quad \dots(i)$$

Replace x by $x+2$, we get

$$f(x) + f(x+2) + f(x+4) = 0 \quad \dots(ii)$$

From (i) and (ii)

$$f(x-2) = f(x+4)$$

Replace x by $x+2$: $f(x) = f(x+6)$

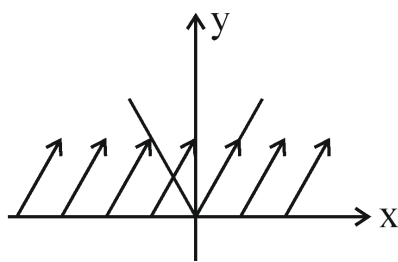
$f(x)$ is periodic with period 6

and $f'(x)$ is also periodic with period 6.

14. **Ans (C)**

$$f(x) = \sqrt{|x| - \{x\}}$$

$$|x| \geq \{x\}$$



$$X \in \left(-\infty, -\frac{1}{2}\right] \cup [0, \infty) \text{ and } Y \in [0, \infty)$$

$Y \in [0, \infty)$ and $f(x)$ is many one

15. **Ans (D)**

$$\sin^{-1}(\cos(\cos^{-1}(\cos x) + \sin^{-1}(\sin x)))$$

$$x \in \left(\frac{\pi}{2}, \pi\right)$$

$$= \sin^{-1}[\cos(x + \pi - x)]$$

$$= \sin^{-1}(-1) = -\frac{\pi}{2}$$

16. **Ans (D)**

$$\text{Rearranging gives : } x^{2020} + (y-1)^2 = 1$$

Clearly x can take values 0, 1, -1 only otherwise $(y-1)^2$ becomes negative putting these values of x we get.

$$R = \{(0, 0), (1, 1), (-1, 1), (0, 2)\}$$

17. **Ans (C)**

$$\text{Shortest distance} = \frac{|(\vec{a} - \vec{b}) \times \vec{d}|}{|\vec{d}|}$$

$$\text{where } \vec{d} = \hat{i} + \hat{j} + \hat{k} \text{ \& } \vec{a} - \vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\therefore \text{shortest distance} = \sqrt{2}$$

18. **Ans (B)**

$$f(x) = x \Rightarrow x^2 - 4x + 3 = 0 \Rightarrow x = 3 \text{ or } 1$$

$\therefore 3 \& 1$ are roots of $f(x) = x$ also.

$$f(f(x)) = x$$

$$\Rightarrow (x^2 - 3x + 3)^2 - 3(x^2 - 3x + 3) + 3 = x$$

$$\Rightarrow x^4 - 6x^3 + 12x^2 - 10x + 3 = 0$$

$\therefore 3 \& 1$ are roots of $f(x) = x$, they are roots of

$f(f(x)) = x$ also and

$$\therefore f(f(x)) - x = (x-3)(x-1)(x-1)^2$$

$$= (x-3)(x-1)^3$$

$\therefore 3, 1, 1, 1$ are solutions of $f(f(x)) = x$

19. **Ans (C)**

Sum of coefficients of terms not containing;

$\{y = 3^6, \text{ put } x = z = w = 1, y = 0\}$ and

Sum of coefficient of terms not containing both

x and $y = 2^6$; $\{\text{put } z = w = 1 \& x = y = 0\}$ so

$$\text{required value} = 3^6 - 2^6$$

20. **Ans (A)**

$$\therefore f(k) = \sum_{r=1}^n (a_r) - a_k \Rightarrow a_k + f(k) = \sum_{r=1}^n a_r$$

$$\text{So, } f(1) + a_1 = \sum_{r=1}^n a_r$$

$$f(2) + a_2 = \sum_{r=1}^n a_r$$

$$\therefore a_1 + f(1) = a_2 + f(2) = a_3 + f(3) = \dots$$

$$= a_n + f(n) = \sum_{r=1}^n a_r$$

Now, given :

$$\frac{1}{a_1}, \frac{1}{a_2}, \dots, \frac{1}{a_n} \rightarrow \text{AP}$$

$$\Rightarrow \frac{a_1 + f(1)}{a_1}, \frac{a_2 + f(2)}{a_2}, \frac{a_3 + f(3)}{a_3},$$

$$\dots \frac{a_n + f(n)}{a_n} \rightarrow \text{AP}$$

$$\Rightarrow \frac{f(1)}{a_1}, \frac{f(2)}{a_2}, \frac{f(3)}{a_3}, \dots, \frac{f(n)}{a_n} \rightarrow \text{AP}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3)**

$$\because \sin^2 x, \frac{\cos^2 x}{8}, \frac{\cos^2 x}{8} > 0 \quad \forall x \in (0, \pi)$$

$$\therefore \frac{f(x)}{3} \geq \left\{ \frac{1}{8^2} \right\}^{\frac{1}{3}} \Rightarrow f(x) \geq \frac{3}{4} = m$$

$$\therefore 4m = 3$$

2. **Ans (3)**

$$\because \text{TAV is } 3x + 4y + k = 0$$

$$\therefore \text{Let axis be } 4x - 3y + \lambda = 0$$

Now, parabola equation can be :

$$(4x - 3y + \lambda)^2 = \mu(3x + 4y + k) \quad \dots(i)$$

Comparing equation (i) with :

$$16x^2 - 24xy + 9y^2 + 14x + 2y + 7 = 0$$

$$\Rightarrow \lambda = 1, \mu = -2, k = 3 \quad \therefore 3.00$$

3. **Ans (33)**

$$\because f(x+2) = f(x-1) \Rightarrow f(x) = f(x+3)$$

$$\therefore \text{period of } f \text{ is } 3 \text{ and } g\left(x + \left[\frac{4x^2 + 13}{x^2 + 3}\right]\right)$$

$$= g\left(x - 2\pi - \sin^{-1}(\sin 7)\right)$$

$$\Rightarrow g(x+4) = g(x-7)$$

$$\therefore g(x) = g(x+11)$$

$$\Rightarrow \text{period of } g \text{ is } 11$$

$$\therefore \text{period of } h(x) \text{ is } 33$$

4. **Ans (21)**

$$\therefore A_3 = \begin{bmatrix} a_1 & b_1 & c_1 \\ 0 & b_2 & c_2 \\ 0 & 0 & c_3 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 1/a_1 & k_1 & k_2 \\ 0 & 1/b_2 & k_3 \\ 0 & 0 & 1/c_3 \end{bmatrix}$$

$$\therefore \frac{1}{10} \sum_{n=1}^5 \text{tr}(A_n^{-1})$$

$$= \frac{1}{10} \{ \text{tr}(A_1^{-1}) + \text{tr}(A_2^{-1}) + \dots + \text{tr}(A_5^{-1}) \}$$

$$= \frac{1}{10} \{ 5.2 + 4.8 + 3.18 + 2.32 + 50 \} = 21$$

5. **Ans (3)**

$$\text{For range, } D = 0 \Rightarrow 4(\sin^{-1} \beta)^2 - 4$$

$$\left\{ \frac{\pi}{\sqrt{2}} + \sin^{-1} \alpha \right\} \left\{ \frac{\pi}{\sqrt{2}} - \sin^{-1} \alpha \right\} = 0$$

$$\therefore (\sin^{-1} \beta)^2 - \frac{\pi^2}{2} + (\sin^{-1} \alpha)^2 = 0$$

$$\therefore (\sin^{-1} \alpha)^2 + (\sin^{-1} \beta)^2 = \frac{\pi^2}{2}$$

$$\therefore \sin^{-1} \alpha = \pm \frac{\pi}{2} \text{ \& } \sin^{-1} \beta = \pm \frac{\pi}{2}$$

$$\therefore \alpha = \pm 1, \beta = \pm 1$$

$$\therefore \alpha^2 + \beta^2 + \alpha^2 \beta^2 = 1 + 1 + 1 = 3$$

6. **Ans (4)**

$$\because (a + \sqrt{2}b \cos x)(a - \sqrt{2}b \cos y) = a^2 - b^2$$

differentiate w.r.t x $\rightarrow$

$$-\sqrt{2}b \sin x (a - \sqrt{2}b \cos y) + \sqrt{2}b \sin y \cdot y'$$

$$(a + \sqrt{2}b \cos x) = 0$$

$$\therefore \text{at } \left(\frac{\pi}{4}, \frac{\pi}{4}\right) \Rightarrow -b(a-b) + by'(a+b) = 0$$

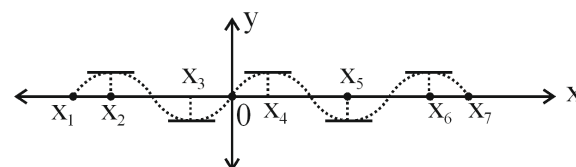
$$\therefore \frac{dy}{dx} = \frac{a-b}{a+b}$$

$$\therefore \frac{dx}{dy} = \frac{a+b}{a-b} = \frac{5+3}{5-3} = 4$$

7. **Ans (10)**

From information given in problem, following

graph can be considered :



$$\therefore f(x) = 0 \text{ has 6 roots and } f'(x) \text{ has 5 roots}$$

$\therefore$ According to graph :

$$g(x) = \frac{d}{dx} \{f(x) \cdot f'(x)\} = 0 \text{ has atleast 10 roots.}$$

8. **Ans (13)**

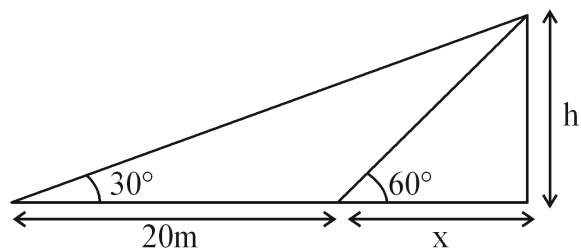
Favourable ways to select 4 squares

$$= 4 \cdot \{ {}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4 \} + 2 \cdot {}^8C_4 = 364$$

$$\therefore \text{required pr} = \frac{364}{64C_4} \therefore \lambda = 364$$

Hence (13)

9. **Ans (10)**



$$\tan 30^\circ = \frac{h}{x+20}, \tan 60^\circ = \frac{h}{x}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{\sqrt{3}x}{x+20} \Rightarrow x = 10$$

10. **Ans (1)**

The series is :

$$1 + 1, (1 + n) + n, (1 + 2n) + n^2, (1 + 3n) + n^3,$$

$$\dots, 1 + 2n + n^2 = (n + 1)^2 \text{ is a perfect square}$$

$$\forall n \in \mathbb{N}$$